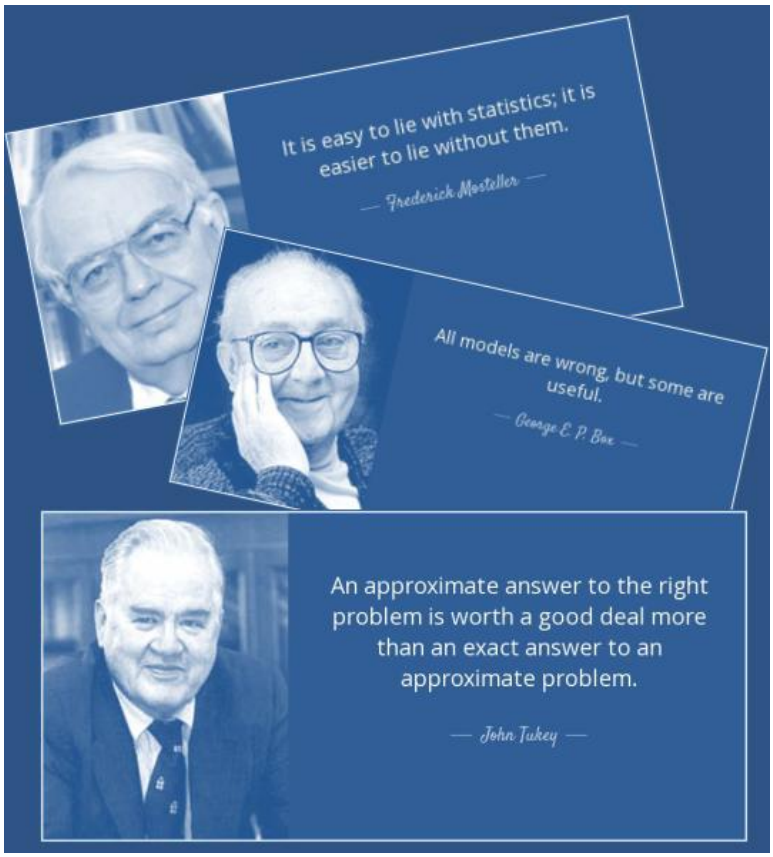
Linear and Longitudinal Data Modeling with R (1 March – 31 July 2026)

Mohammad Arashi – FUM & UP - Preliminary notes

- This is an online course! You cannot ask your questions while I am teaching, but I try to provide you with several examples that you can understand.
- Some sessions may go for 2-3 weeks.
- The planner includes 28 videos for 14 weeks (not Calander weeks) which will be expanded into 5 months. Some sessions are live for Q&A.
- Remember successive sessions are chained together; don't miss one.
- The assignments are mandatory but the "Why" questions are FYI.
- We will have R laboratories (Lab) within the sessions for the R codes.
- Use the integrated development environment for R, the Rstudio available at https://rstudio.com/ or https://posit.co/ for coding.
- During Q&A, we will work with Posit Cloud.



Curriculum: Our agenda

1- Fixed-effects regression (review of regression)
2- Mixed-effects regression (including random effect)
3- Longitudinal data analysis (allowing correlated data)
4- Generalized linear models (with focus on longitudinal model)
5- Penalized linear model (we will master estimation in linear models)
6- Non-linear regression (non-linear footprint)

Main reference

- Verbeke, G. and Molenberghs, G. (2001) Linear Mixed Models for Longitudinal Data, Springer, New York.

Recommended reference

- Galecki, A. and Burzykowski, T. (2013) Linear Mixed-Effects Models Using R: A Step-by-Step Approach, Springer, New York.

Planner

	Chapter	Subject	Lecture Note	Video Structure Guide	Practical
1-31 March	#1	Fixed-effects & Mixed-effects regressions	Sessions 1 & 2	From: Week1-part1 To: Week4-part2	#1 & 2
1-30 April	#2	Longitudinal data analysis	Session 3	From: Week5-part1 To: Week7-part2	#3
1-31 May	#3	Generalized linear models	Session 4	From: Week8-part1 To: Week9-part2	#4
1-30 June	#4	Penalized linear model	Session 5	From: Week10-part1 To: Week13-part1	#5
1-31 July	#5	Non-linear regression	Session 6	From: Week13-part2 To: Week14-part2	